Introduction to psycholinguistics

Day 2: Language, communication, animal communication

fixed and pruned down version!

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Stuttgart
Oct. 26th, 2026

Last time

Last week, we saw two experiments. These both were perception experiments, as opposed to production experiments. One demonstrated a phonetic/phonological phenomenon and the other, a syntactic/semantic phenomenon.

We drew connections with vision, dresses, slurs and Taylor Swift

Last week, we saw two experiments. These both were perception experiments, as opposed to production experiments. One demonstrated a phonetic/phonological phenomenon and the other, a syntactic/semantic phenomenon.

We drew connections with vision, dresses, slurs and Taylor Swift (it *is* all connected).

These experiments had to do with first language use in adults as opposed to second language use or language acquisition in infants. Modulo, yes, the fact that we are most second language users of English ;)

In these experiments, you were **participants**. These are conducted to answer questions, about language in our case, which are cast and re-cast as **hypotheses**.

In these experiments, you were **participants**. These are conducted to answer questions, about language in our case, which are cast and re-cast as **hypotheses**. They are subject to **methodological** rigor.

My goal was to suggest that language use has effects on the mind and body that are not obvious, and ones that can be studied experimentally (among other ways).

I always hedge when I say ‘mind’, ‘body’, and ‘mind and body’ because that all corresponds to a big philosophical debate that is… delicate.

That this all has something to do with the intersection between what constitutes language and its properties (linguistics),

My goal was to suggest that language use has effects on the mind and body that are not obvious, and ones that can be studied experimentally (among other ways).

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That this all has something to do with the intersection between what constitutes language and its properties (linguistics), and how it is produced, perceived, acquired, mind or brain-wise represented etc. in the individual (psychology).

A metaphor about scientific inquiry?

Uzun ince bir yoldayım,	*I am on a long and narrow road,*
Gidiyorum gündüz gece,	*I walk day and night;*
Bilmiyorum ne haldeyim,	*I do not know what state I am in*
Gidiyorum gündüz gece.	*I walk day and night;*
Dünyaya geldiğim anda,	*The moment I came into the world,*
Yürüdüm aynı zamanda,	*I walked at the same time*
İki kapılı bir handa	*At an inn with two doors*
Gidiyorum gündüz gece.	*I walk day and night.*

an original version (Aşık Veysel):
`https://youtu.be/_ykio03RSRc`

a modern (the nineties?) interpretation (Barış Manço and Cem Karaca):
`https://youtu.be/iibPxAxlD64`

a contemporary version (ENGIN):
`https://youtu.be/xAM4y8K4yhI`

Today

Let’s try to define language and see if it’s relevant at all to try to do so, and talk about the birds and the bees.

What is language?

What is language?

What is language?
“It’s a system of communication.”

What is language?

~~“It’s a system of communication.”~~

What is love?
It's a feeling.

What is a dog?
It's an animal.

What is language?
It's a system of communication.

- These answers are not specific enough.

  What distinguishes love from other feelings, dogs from other animals, language from other systems of communication?

- Is it true (or important) that language is a system of communication?

Quick fix: Add more qualifiers.

What is language?
It's a system of communication

That's a bit better, but...

- There have to be enough conditions on the right.
- There can't be too many conditions on the right.
- The conditions on the right have to be true.

Quick fix: Add more qualifiers.

What is language?
It's a system of communication used by all humans

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It's a system of communication used by all humans and only by humans

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It's a system of communication used by all humans and only by humans, instinctively

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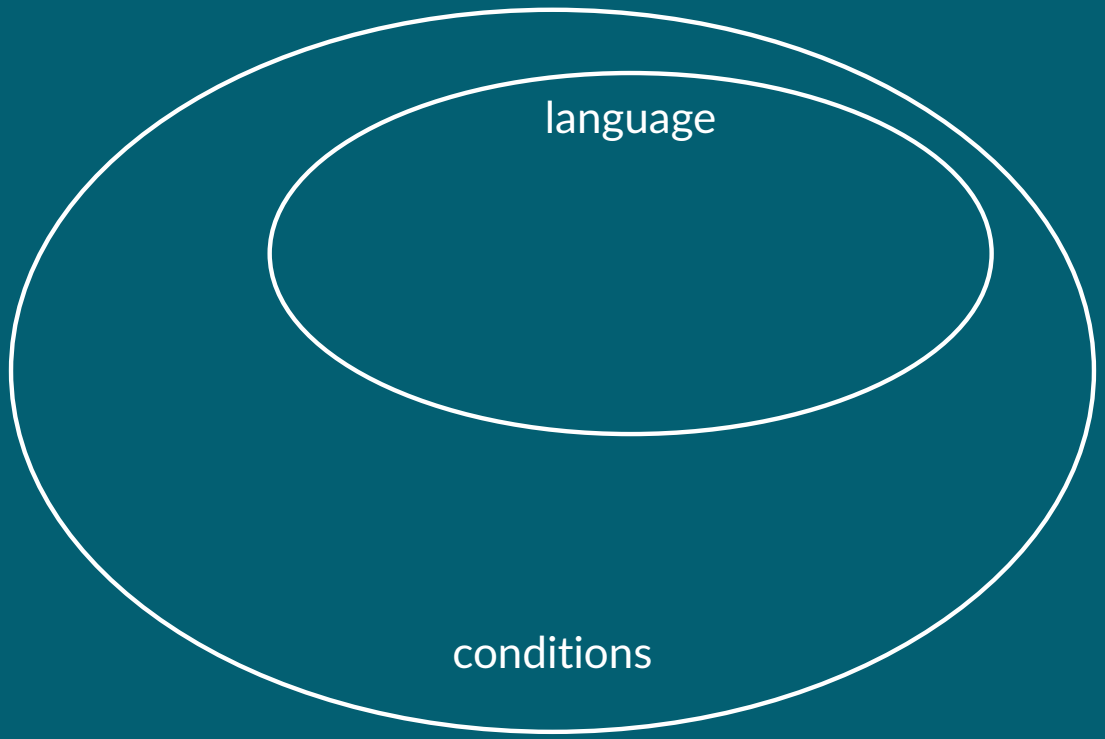
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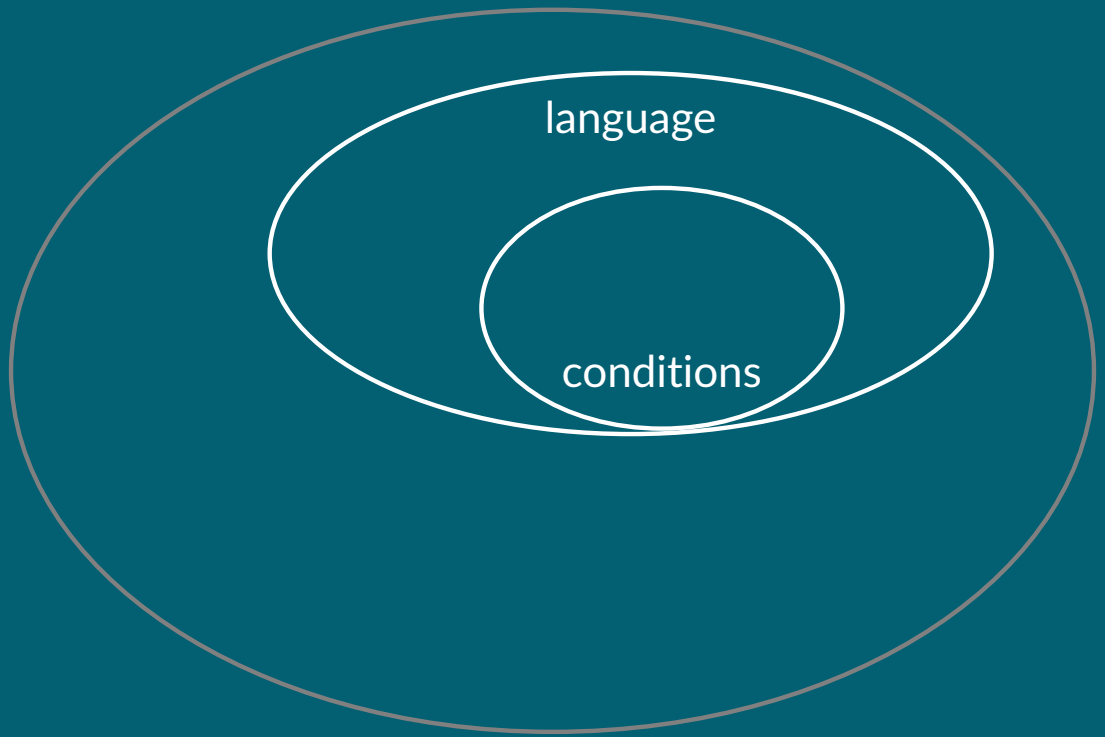
The problem of too few conditions



The ancient Greeks debated this topic feverishly. Plato famously attempted to define a man, using references from his mentor Socrates. He settled on a scientific definition, naming man a "featherless biped," two characteristics that distinguished humanity from other animals. In a humorous scene, Diogenes the Cynic, hearing Plato's definition, plucked a chicken in his home and brought it to one of Plato's lectures. When Plato asserted that man was a featherless biped, Diogenes stood, brandished the bald chicken and shouted, "Behold—a man!" Plato, perhaps missing the point of Diogenes' criticism, then continued to amend his definition to "featherless biped with flat, broad nails."

This formulation of the anecdote taken from `here`.

The problem of too many conditions



Definitions

First published Thu Apr 10, 2008; substantive revision Wed Sep 13, 2023

Definitions have interested philosophers since ancient times. Plato's early dialogues portray Socrates raising questions about definitions (e.g., in the *Euthyphro*, "What is piety?")—questions that seem at once profound and elusive. The key step in Anselm's "Ontological Proof" for the existence of God is the definition of "God," and the same holds of Descartes's version of the argument in his *Meditation V*. More recently, the Frege-Russell definition of number and Tarski's definition of truth have exercised a formative influence on a wide range of contemporary philosophical debates. In all these cases—and many others can be cited—not only have particular definitions been debated; the nature of, and demands on, definitions have also been debated. Some of these debates can be settled by making requisite distinctions, for definitions are not all of one kind: definitions serve a variety of functions, and their general character varies with function. Some other debates, however, are not so easily settled, as they involve contentious philosophical ideas such as essence, concept, and meaning.

- 1. Some varieties of definition
  - 1.1 Real and nominal definitions
  - 1.2 Dictionary definitions
  - 1.3 Stipulative definitions
  - 1.4 Descriptive definitions
  - 1.5 Explicative definitions
  - 1.6 Ostensive definitions
  - 1.7 A remark
- 2. The logic of definitions
  - 2.1 Two criteria
  - 2.2 Foundations of the traditional account
  - 2.3 Conservativeness and eliminability
  - 2.4 Definitions in normal form
  - 2.5 Conditional definitions
  - 2.6 Implicit definitions
  - 2.7 Vicious-Circle Principle
  - 2.8 Circular definitions
- Bibliography
- Academic Tools
- Other Internet Resources
- Related Entries

Stanford Encyclopedia of Philosophy entry

Back to communication

"Language is a system of communication" invites the following question (to me, at least).

True or false (discuss):

- "Every time we use language, we communicate something
- and every time we communicate, we use language."

Back to humans vs. non-human animals

I don't know of anybody who would deny that all humans have language or are capable of it.

There are fringe cases, but they don't cast a doubt on this claim.

- the Pirahã controversy
- traumatic events (e.g., Genie) or atypical development

Back to humans vs. non-human animals

Whether only humans have language is less clear.

- Bees dance to communicate.*
  Their dance is very different from our linguistic behavior.
- Some non-human primates' vocalizations might come closer to what we have.
- There's much to say about animal communication, we'll talk a bit about that next week.

*I think people say "bee communication" to avoid saying "bee language," but if you agree that language $\neq$ communication (in humans), the comparison here has to be more careful.

Language isn't dependent on modality

All have shared properties:

- spoken languages
- sign languages
- tactile languages

[demo]

What is and isn't language depends on

- (formal) properties that are common to all particular languages

  units (sounds, words) and ways of combining those units into bigger chunks (structure building)

- properties of these structures

  e.g., recursion, linguistic universals

- biological or psychological properties

  babies acquire language effortlessly, instinctively, etc.

And there's a lot that we don't know!

This course will be an opportunity to learn about all of this!

What (psycho)linguistics do?

Linguists try to understand how language works and how languages work.

- ✧ It's hard to define language and what counts as a *particular* language in a way that is both concise and exact.
- ✧ But it's relatively easy to describe…
  - what linguists (and psycholinguists) do,
  - the kinds of phenomena that they study,
  - the methods that they use,
  - and what they learn about language(s).
- ✧ Let's review what linguists and psycholinguists do.

Situating psycholinguistics

Linguistic subfields

Linguistics breaks down into different ‘subfields.’

Phonetics
How sounds are physically produced and perceived.

Phonology
How sounds are categorized and treated abstracting away from the physical signal.

Morphology
How words are internally structured.

Situating psycholinguistics

Linguistic subfields

Linguistics breaks down into different ‘subfields.’

Syntax
How phrases and sentences are internally structured.

Semantics
What the literal meanings of phrases and sentences are.

Pragmatics
What is the speaker’s intent is in saying something, beyond its literal meaning.

The same/analogous notions exist for sign and tactile languages.

Situating psycholinguistics

Linguists differ in how they approach similar phenomena.

They employ different tools, work in different places, formulate different kinds of hypotheses…

You could be…

- a theoretical syntactician,
- a fieldworker and a syntactician,
- a computational linguist and a syntactician,
- a psycholinguist and a syntactician,
- a sociolinguist and a syntactician,
- …

Most linguists do more than one thing, with the common goal of understanding how language works and how languages work.

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"[Psycholinguistics is] a field that uses experimental methods to study the psychological machinery that drives language learning, language comprehension, and language production."

Sedivy (2019)

Recap

- The relationship between language and communication.
  - All language is communication (unclear, maybe true)
  - All communication is language (clearly false)

- The relationship between language and communication.
  - All language is communication (unclear, maybe true)
  - All communication is language (clearly false)
- The relationship between having language and being human.
  - All humans have language (clearly true)
  - Only humans have language (unclear, probably false)

Next steps

- Some more on the general notion of a language
- Honey bee and non-human primate communication.
  - What kinds of experiments, what kinds of findings?
  - Do these means of communication share properties with human languages?*

*One can ask this question without committing to any answer to whether or not human language = communication.

More on human language(s)

Until the section on honey bees, we only looked, in class, at a couple of Greenberg's universals and the exchange between ADULT and CHILD. I leave the rest in if you're curious.

Units and their combination

Languages have

- a set of basic units,
- a generative component to combine them into complex units.

Units and their combination

In human languages, there are at least two sets of basic units:

- a set of sounds (phonemes):
  /p/, /t/, /k/, /æ/, /n/, …
- a set of (parts of) words (morphemes):
  un-, lock, -able, dinosaur, …

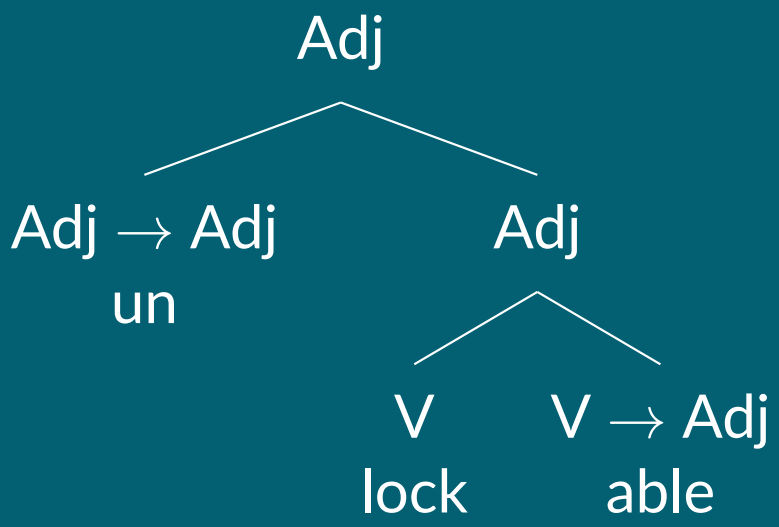
Units and their combination

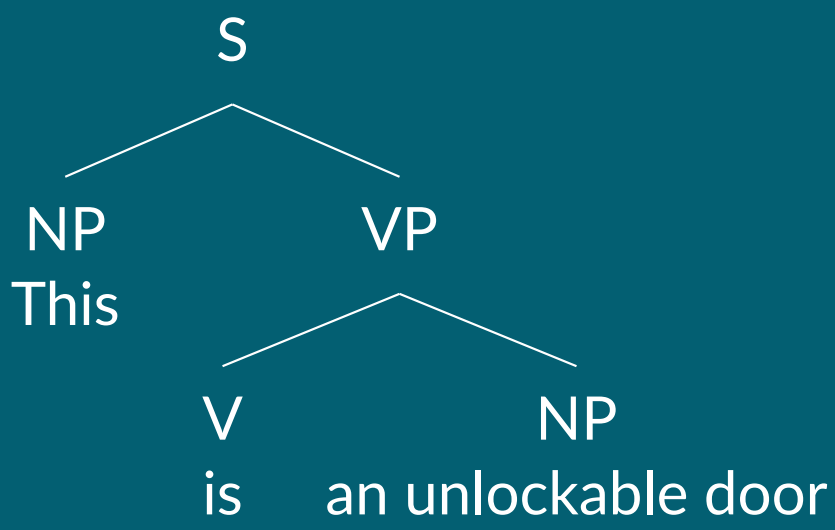
The generative component for sounds is (roughly) concatenation:

/pæn/, /næp/, …

Units and their combination

For morphemes & words, it's morphosyntactic structure building:





Combinations are governed by rule

Not every combination is possible: These generative components are governed by rules and constraints.

(1) *[pnæ], *[npæ], …

(2) *This an is unlockable door.

Some of these are language specific.

(3) a. …weil du gegessen hast.
*…weil du hast gegessen.

b. …because you have eaten.
*…because you eaten have.

Some are general to all languages.

Generalizations about the systems

By studying large numbers of languages, one can try to make generalizations about all or large groups of them.

Explore properties of large groups of languages for yourself:

- World Atlas of Language Structures (WALS)

Other databases record variation within a language:

- Syntax of Alemannic (SynAlm)
- ScotsAtlas
- Yale Grammatical Diversity Project

...

Language universals?

Greenberg (1966)

CHAPTER 5

SOME UNIVERSALS OF GRAMMAR WITH PARTICULAR REFERENCE TO THE ORDER OF MEANINGFUL ELEMENTS

JOSEPH H. GREENBERG

Stanford University

1. Introduction

The tentative nature of the conclusions set forth here should be evident to the reader. Without much more complete sampling of the world's languages, the absence of exceptions to most of the universals asserted here cannot be fully assured. As indicated by the title, attention has been concentrated largely, but by no means exclusively, on questions concerning morpheme and word order. The reason for this choice was that previous experience

Language universals?

Greenberg (1966)

Universal 1. In declarative sentences with nominal subject and object, the dominant order is almost always one in which the subject precedes the object.[5]

Language universals?

Greenberg (1966)

Universal 4. With overwhelmingly greater than chance frequency, languages with normal SOV order are postpositional.

Language universals?

Greenberg (1966)

Universal 8. When a yes-no question is differentiated from the corresponding assertion by an intonational pattern, the distinctive intonational features of each of these patterns are reckoned from the end of the sentence rather than from the beginning.

Another important property: Recursion

All human languages are capable of recursion:

(4) a. the student

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(4) a. the student
b. the student next to the window

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c. the student next to the window in KI

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d. the student next to the window in KI on Kepler Str.
…

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All human languages are capable of recursion:

(4) a. the student
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(5) a. I said let's go.

Another important property: Recursion

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(5)
a. I said let's go.
b. You think that I said let's go.

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…

(5)
a. I said let's go.
b. You think that I said let's go.
c. Zoe knows that you think that I said let's go.

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a. the student
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Ungrammatical examples

What is the rule for forming yes/no questions in English?

(6) a. The man who is tall is happy.

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b. Is the man who is tall happy?

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If you want to convince me that you *have to* move the main auxiliary to the beginning of the sentence, you *also* have to show me (6c).

Poverty of the stimulus

Part of the mystery of language acquisition is that children don't make mistakes like (6c).

(5) c *Is the man who tall is happy?

Yet… nobody has told them that this is not a good sentence of English. They somehow learn the English question formation rule on the basis of incomplete evidence.

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Yet… nobody has told them that this is not a good sentence of English. They somehow learn the English question formation rule on the basis of incomplete evidence.

This is called the **poverty of the stimulus**. People like Noam Chomsky have used this observation to argue that there must already be something in our brains/minds that constrains what is or isn't possible in (the) language (that we're acquiring).

They do make mistakes, that suggest that they are acquiring rules and that they resist correction.

```
CHILD: My teacher holded the baby rabbits and we patted them.
ADULT: Did you say your teacher held the baby rabbits.
CHILD: Yes.
ADULT: What did you say she did?
CHILD: She holded the baby rabbits and we patted them.
ADULT: Did you say she held them tightly?
CHILD: No, she holded them loosely.
```

What does this exchange tell us about how children acquire language? Can you also conclude something about example (6)?

Wrapping up

When lay people talk about language, I think they have this in mind (a set of basic units + a generative component) even though they might not say things in those terms.

Another thing that they have in mind is *expressivity* or *communication*.

These notions are found in: “body language, the language of design, love language, the language of the bee,” etc.

Honey bees

Honey bee dances

DECODING THE LANGUAGE OF THE BEE

Nobel Lecture, December 12, 1973

by

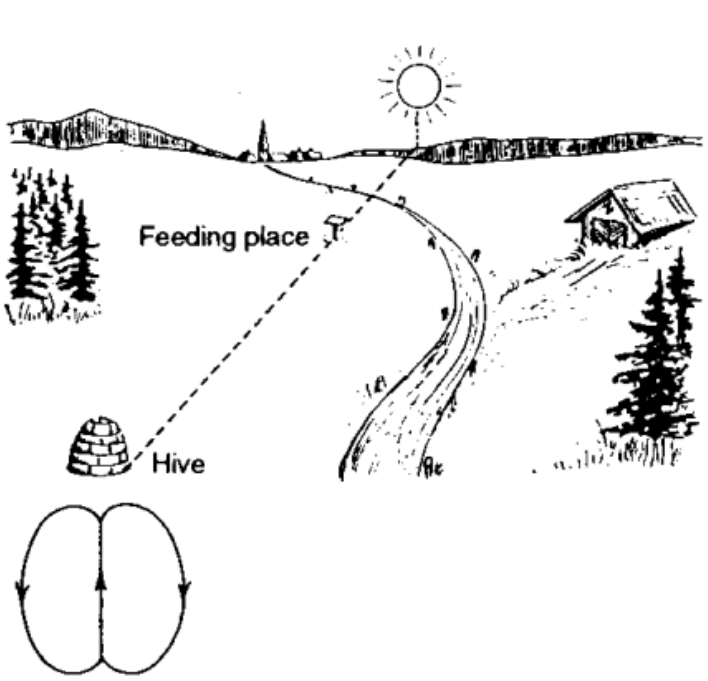
KARL VON FRISCH†

University of Munich, Federal Republic of Germany

Some 60 years ago, many biologists thought that bees and other insects were totally color-blind animals. I was unable to believe it. For the bright colors of flowers can be understood only as an adaptation to color-sensitive visitors. This was the beginning of experiments on the color sense of the bee (I). On a table outdoors I placed a colored paper between papers of different shades of gray and on it I laid a small glass dish filled with sugar syrup. Bees from a nearby hive could be trained to recognize this color and demonstrated their ability to distinguish it from shades of gray. To prevent too great a gathering of bees, I instituted breaks between feedings. After these breaks, only sporadic

- ✧ What are the main empirical findings of this paper?
- ✧ What factors does von Frisch manipulate in his experiments?
- ✧ What makes us think that it *is* properties of their dance that help bees locate food, rather than, e.g., scent?
- ✧ What similarities and differences do you see between bees' dances and human language(s)?

Honey bees



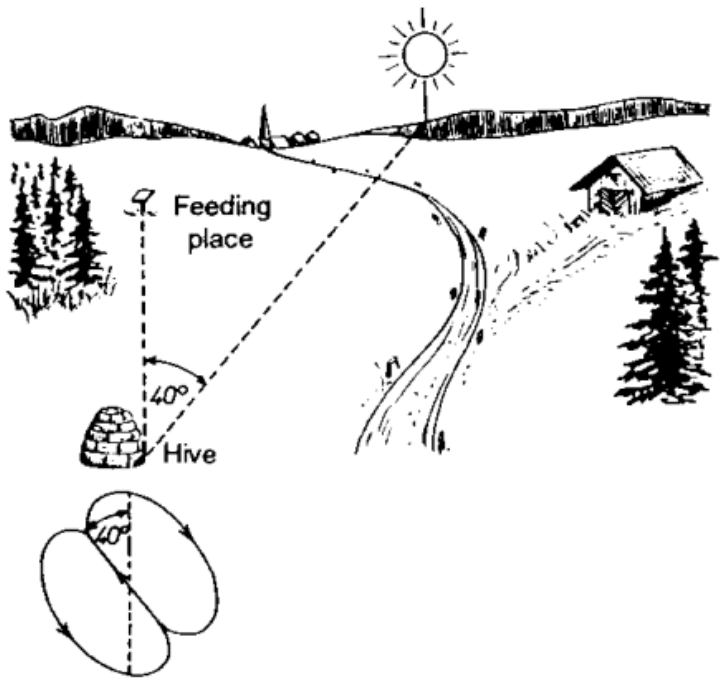


Fig. 3. Indication of direction by tail-wagging dance. (Left), the goal is in the direction of the sun; (right), the goal is 40° to the left of the sun's position. Dance figures, enlarged, are on the bottom left of the pictures.

Honey bees

Their dance encodes:

- The direction of nectar source
  Angle of deviation from the vertical = angle relative to the sun.
- The distance to the source
  The slower the dance, the farther away the source.
- The quality of the source
  The more intense the waggles, the better the source.

Demo: https://youtu.be/PIkNqLMFGA0

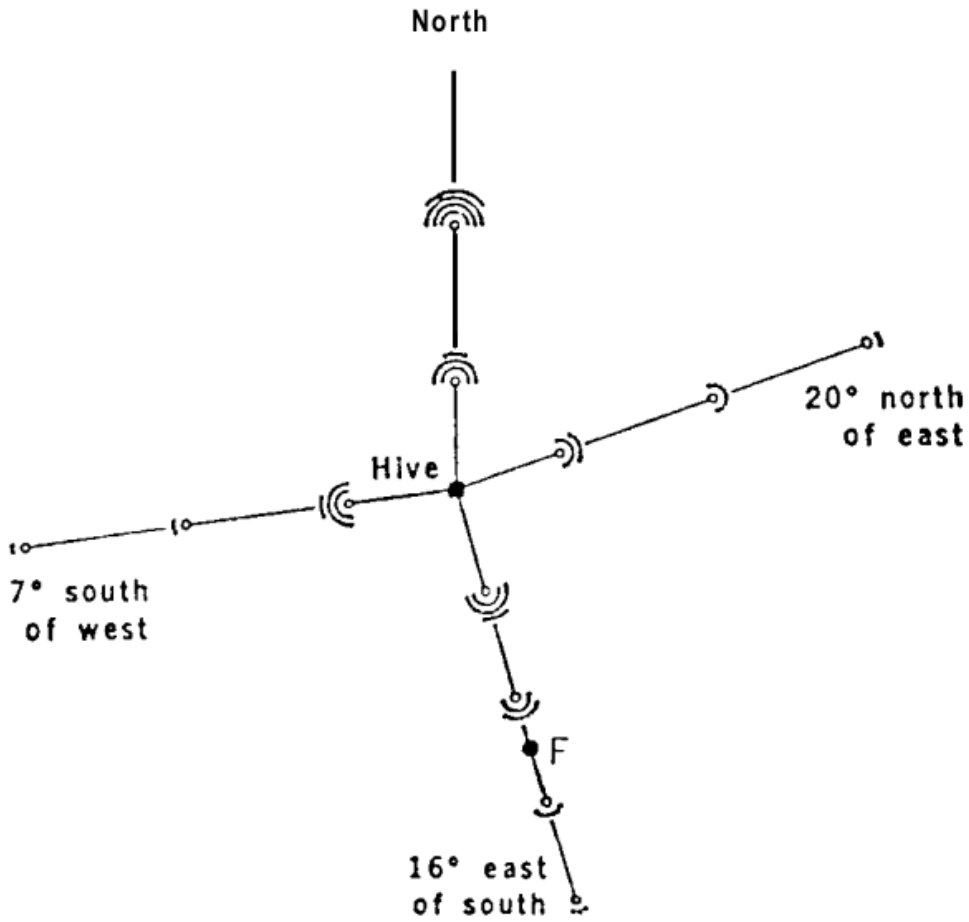


Fig. 6. Effect of placing observation hive horizontally. The dances are disoriented. Scented plates with the scent of the feeding place are visited by great numbers of forager recruits (small dots) in all few directions: F, feeding place.

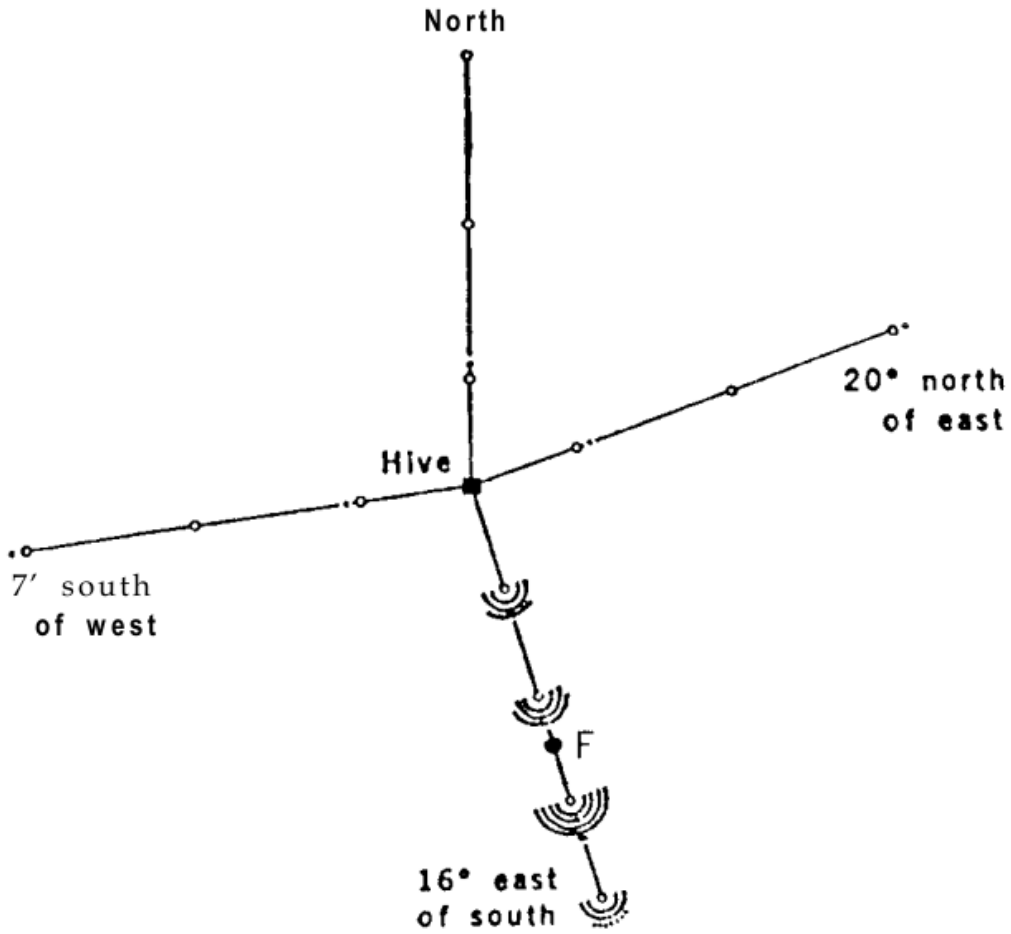


Fig. 7. Hive placed vertically after experiment in Fig. 6. The dances now indicate the direction of the feeding place. Within 10 minutes the stream of forager recruits turns in this direction. Flights no longer arrive in the three other directions.

Non-human primates

we were not able to get to this on Day 2

Vervet monkey alarm calls

Anim. Behav., 1980, **28**, 1070–1094

VERVET MONKEY ALARM CALLS: SEMANTIC COMMUNICATION IN A FREE-RANGING PRIMATE

BY ROBERT M. SEYFARTH, DOROTHY L. CHENEY & PETER MARLER
Rockefeller University, Field Research Center, Millbrook, N.Y. 12545

Abstract. Vervet monkeys (*Cercopithecus aethiops*) at Amboseli, Kenya, give acoustically different alarm calls to different predators. Each alarm evokes contrasting, seemingly adaptive, responses. Animals on the ground respond to leopard alarms by running into trees, to eagle alarms by looking up, and to snake alarms by looking down. In a 14-month field study examining the semantic properties of alarm calls, we played tape-recorded alarms to vervets in the absence of actual predators and filmed the monkeys' responses. Playbacks confirmed observations and showed that (1) alarm length, amplitude and alarmist's age/sex class had little effect on response quality, and (2) context was not a systematic determinant of response. We conclude that vervet alarm calls function to designate different classes of external danger.

Demo: https://www.youtube.com/watch?v=3lsF83rHKFc

- What are the main empirical findings of this paper?
- What are the two kinds of methods that the authors use to substantiate their findings?
- What linguistic properties do Vervet monkey alarm calls seem to have, that human languages also have?
- What linguistic properties do Vervet monkey alarm calls *not* seem to have?

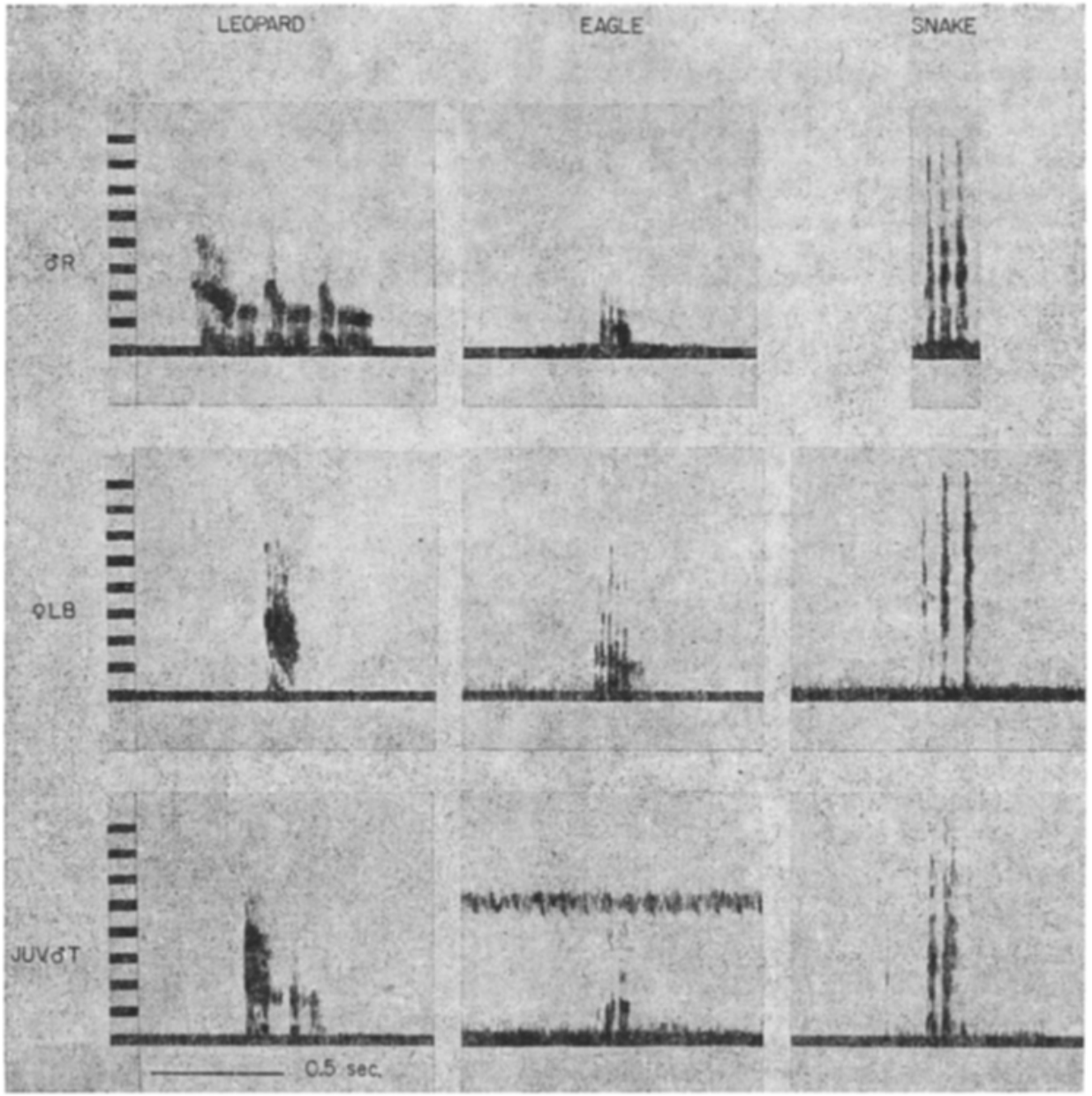


Fig. 1. Examples of the alarms given by three individuals to leopards, martial eagles, and snakes. Top row shows alarms by adult male R (Group C); middle row shows alarms by adult female LB (Group A); and bottom row shows alarms by 2.5 year-old juvenile male T (Group C). X-axis shows time; Y-axis shows frequency in units of 1 kHz. In eagle alarms of juvenile T, noise around 5 kHz is insects.

Part I: Observation

Table IV. The Number of Occasions When at Least One Individual Showed a Particular Response to a Given Type of Alarm

Ground	Responses				
Alarm type	Run into tree	Run into cover	Look up	Look down	Approach predator
Leopard $N = 29$	20	1			
Eagle $N = 108$	6	26	64		
Snake $N = 16$	1	2		8	7

Tree	Responses				
Alarm type	Run higher in tree	Run out of tree	Look up	Look down	Approach predator
Leopard $N = 24$	2				
Eagle $N = 33$	1	2	10		
Snake $N = 10$				8	9

N = Number of occasions when each type of alarm was given as a first alarm. Upper table shows responses when monkeys were on the ground; lower table shows responses when monkeys were in a tree.

Part II: Playback experiments

Table VII. The Number of Trials involving Playback of Different Alarm Types in Which at Least One Animal showed a Given Response for Longer in the 10 s After Playback than in the 10 s Before Playback

Ground	Responses						
Alarm	Look toward Speaker	Scan	Run into Tree	Run into Cover	Look Up	Stand Bipedally	Look Down
Leopard (19 trials)	10	10	8	2	4	9	1
Eagle (14 trials)	12	11	2	6	7	6	4
Snake (19 trials)	19	19	2	2	2	12	14

Tree	Responses					
Alarm	Look toward Speaker	Scan	Run Higher in Tree	Run Out of Tree	Look Up	Look Down
Leopard (10 trials)	9	8	4	0	3	4
Eagle (17 trials)	16	14	4	5	11	12
Snake (9 trials)	5	6	2	0	5	9

Grammar in non-human primates?

Primate calls

Monkey Semantics:

Two 'Dialects' of Campbell's Monkey Alarm Calls*

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To appear in *Linguistics & Philosophy*

Abstract. We develop a formal semantic analysis of the alarm calls used by Campbell's monkeys in the Tai forest (Ivory Coast) and on Tiwai island (Sierra Leone) – two sites that differ in the main predators that the monkeys are exposed to (eagles on Tiwai vs. eagles and leopards in Tai). Building on data discussed in Ouattara et al. 2009a,b and Arnold et al. 2013, we argue that on both sites alarm calls include the roots *krak*, and *hok*, which can optionally be affixed with *-oo*, a kind of attenuating suffix; in addition, sentences can start with *boom boom*, which indicates that the context is not one of predation. In line with Arnold et al. 2013, we show that the meaning of the roots is not quite the same in Tai and on Tiwai: *krak* often functions as a leopard alarm call in Tai, but as a general alarm call on Tiwai. We develop models based on a compositional semantics in which concatenation is interpreted as conjunction, roots have lexical meanings, *-oo* is an attenuating suffix, and an all-purpose alarm parameter is raised with each individual call. The first model accounts for the difference between Tai and Tiwai by way of different lexical entries for *krak*. The second model gives the same underspecified entry to *krak* in both locations (= general alarm call), but it makes use of a competition mechanism akin to scalar implicatures. In Tai, strengthening yields a meaning equivalent to *non-aerial dangerous predator* and turns out to single out leopards. On Tiwai, strengthening yields a nearly contradictory meaning due to the absence of ground predators, and only the unstrengthened meaning is used.

Morphosyntactic complexity

(20) **Informal description of the lexical meanings** (except *wak*, *wak-oo*)
a. boom boom: 'this is not a situation of predation'
b. krak-oo: 'there is an alert'
c. hok: 'there is an eagle'
d. hok-oo: 'there is an alert upwards'
e. krak: (i) 'there is a leopard' (Tai); (ii) 'there is an alert' (Tiwai)

Morphosyntactic complexity

(6) **Abbreviations**
boom = b, krak = k, krak-oo = K, hok = h, hok-oo = H, wak = w, wak-oo = W

(7) **Examples from the Tai forest [new dataset, playback experiments]**
a1. Eagle shrieks
(i) hhhk
(ii)hhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhWhhhhWWhHWhhWWhWhhWhWh
WWwhWWWwWhhKhKKKWKWKKKhhKWWKKKKKKKKKKKKKKKKKKKKKKK

a2. Campbell eagle call:
(i) hhhhhhhhhhhhHhhhhhHhHHHWWWWHHhHWWWWWWWWWWWWWWWWWWW
(ii) KKKKKKKK

b1. Leopards growls
(i) kkkkkkKK
(ii) kkkkkkkkkkkkkkkkkkk

Morphosyntactic complexity

(10) **Sample syntactic rules (non-exhaustive)**

a. Cohesion and Travel:	bb
b. Tree/branch:	bb K*
c. Inter-group:	bb (K) H * K *
d. Leopard (various contexts):	k K*, k k*
e. Eagle (various contexts):	K K*, h*k, K*h*K*

Pragmatics

In the Tai forest (Côte d'Ivoire), the call *krak* warns against leopards & ground predators. On Tiwai island (Sierra Leone), the same call is used as a general warning call.

Both sites have the call *hok* for aerial predators.

Tai has eagles and leopards. Tiwai has eagles but no leopards.

Pragmatics

krak: there is a disturbance
krakoo: there's a weak disturbance
hok: there's an aerial predator